

Grade 6 Accelerated
Unit 7
Lesson 5 Homework

Name Answer Key
Date _____
Period _____

1. The formula to find the area of a parallelogram is

$$A = b \cdot h$$

2. The formula to find the area of a trapezoid is

$$A = \frac{1}{2} \cdot h \cdot (b_1 + b_2)$$

3. The formula to find the area of a rhombus is

$$A = \frac{1}{2} \cdot d_1 \cdot d_2$$

For questions 4-6, label each figure with the given dimensions, then calculate the area using the appropriate formula.

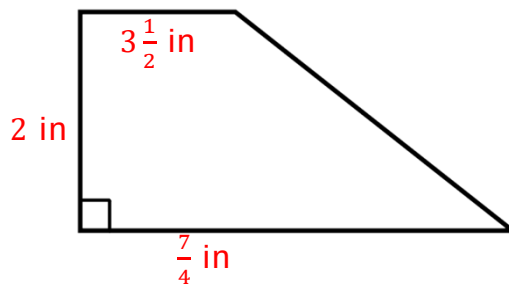
4. $b = 12.3$ in
 $h = 7.1$ in
 $12.3 \cdot 7.1$

$$\text{Area} = 87.33 \text{ in}^2$$



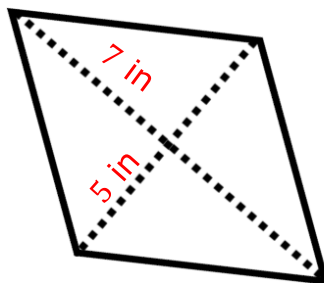
5. $b_1 = 3\frac{1}{2}$ in
 $b_2 = \frac{7}{4}$ in
 $h = 2$ in
 $\frac{1}{2} \cdot 2 \cdot \left(3\frac{1}{2} + \frac{7}{4}\right)$

$$\text{Area} = 5\frac{1}{4} \text{ in}^2$$



6. $d_1 = 7$ in
 $d_2 = 5$ in
 $\frac{1}{2} \cdot 7 \cdot 5$

$$\text{Area} = 17.5 \text{ in}^2$$



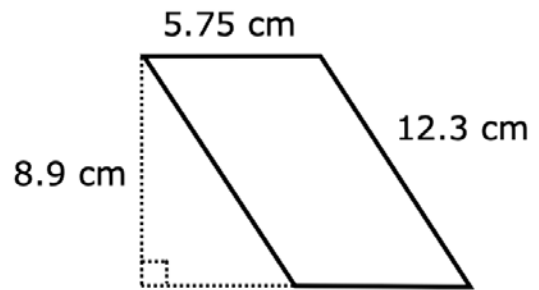
7. Zoey made an error in her work. Identify the error and then find the area of the parallelogram.

$$\text{Area} = (5.75)(12.3)$$

$$\text{Area} = 70.725 \text{ cm}^2$$

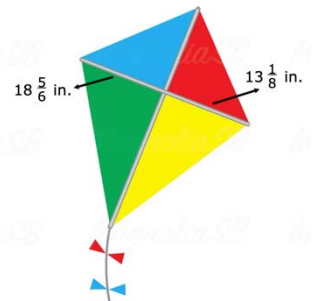
Zoey did not use the actual height; she used the slant height.

$$A = 5.75(8.9) = 51.175 \text{ cm}^2$$

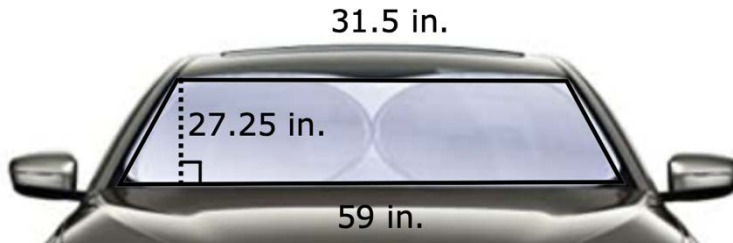


8. Kai and his father are constructing a kite to fly at the beach that is in the shape of a rhombus. The spars on the kite measure $13\frac{1}{8}$ inches and $18\frac{5}{6}$ inches. How many square inches (in^2) of fabric will be needed to construct the kite?

$$A = \frac{1}{2} \left(13\frac{1}{8} \right) \left(18\frac{5}{6} \right) = 123\frac{19}{32} \text{ in}^2$$



9. The windshield of a car is in the shape of a trapezoid. Find the area of the glass used to make the windshield.



$$A = \frac{1}{2} \cdot 27.25 \cdot (31.5 + 59)$$

$$= \frac{1}{2} \cdot 27.25 \cdot 90.5$$

$$= 1233.0625 \text{ in}^2$$

10. Marc needs a new roof on his house and needs to measure how many roofing tiles are needed to get an accurate estimate from the roofing company. The roof has two sides. Each side is in the shape of a parallelogram with a base of $56\frac{5}{12}$ feet and a height of $28\frac{1}{2}$ feet. Find the how many square feet of roofing material is needed.

$$\text{Area} = 56\frac{5}{12} \left(28\frac{1}{2} \right) = 1607\frac{7}{8} \text{ or } 1607.875$$

$$\text{Total area} = 2(1607.875) = 3215.75 \text{ or } 3215\frac{3}{4} \text{ ft}^2$$

